

FREIGHT COST PREDICTION & MODEL DEPLOYMENT

End-to-End Supervised Regression Pipeline, Model Benchmarking & Streamlit Deployment

Live Streamlit Application: <https://adsfa2-machine-learning.streamlit.app/>

GitHub Project Repository: <https://github.com/NileshJadhav1312/ADSFA2>

Google Drive Dataset (inventory.db): <https://drive.google.com/file/d/1o6S9N0j77qM4fd9kbdE2gw2V1TVeh5Ey/view?usp=sharing>

CANDIDATE MODEL PERFORMANCE BENCHMARK

Model Name	MAE (\$)	MSE	RMSE (\$)	R ² Score	Performance Ranking
Random Forest (Tuned via GridSearchCV)	\$6.97	550.45	\$23.46	0.9866	1st (Selected Best Model)
Linear Regression (OLS Baseline)	\$6.85	561.83	\$23.70	0.9863	2nd
Gradient Boosting Regressor	\$7.67	569.08	\$23.86	0.9861	3rd
Random Forest (Default)	\$7.50	631.72	\$25.13	0.9846	4th
Decision Tree Regressor	\$8.14	927.16	\$30.45	0.9774	5th

Selected Best Model: Random Forest Regressor (Tuned) achieved the lowest overall test error (**RMSE = \$23.46**) and highest goodness of fit (**R² = 0.9866**), explaining 98.66% of actual shipping cost variance. Hyperparameters identified: max_depth=6, min_samples_split=5, n_estimators=50.

MODEL EVALUATION & STREAMLIT INTERFACE VISUALIZATIONS

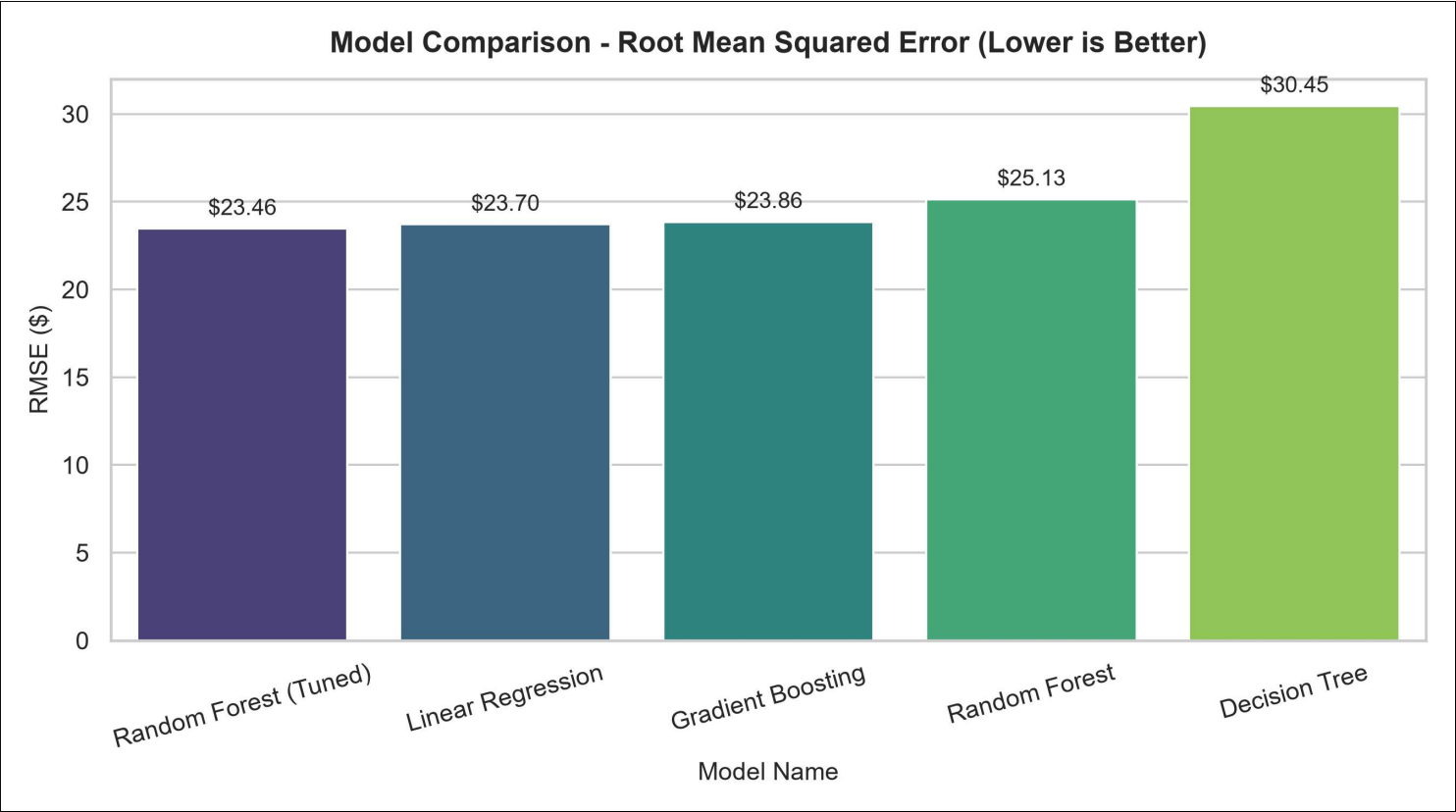


Figure 1: RMSE Comparison across Candidate Models (Lower is Better)

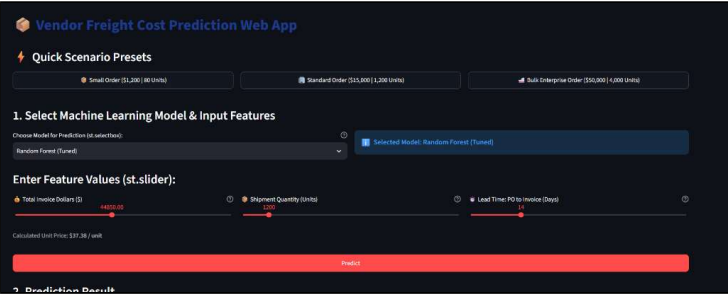


Figure 3: Streamlit Interface — Scenario Presets, Model Selector & Sliders

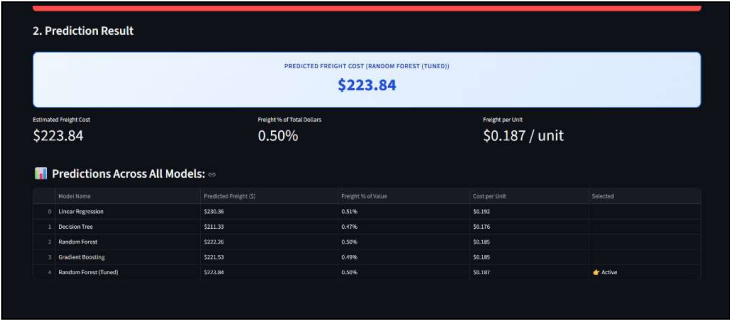


Figure 4: Real-Time Prediction Output (\$223.84 for \$44.8k Invoice) & Multi-Model Table

KEY FINDINGS, PIPELINE LEARNINGS & INSIGHTS

Part 1 Insights: Data Acquisition, Exploration & Preprocessing

- **Dominance of Invoice Monetary Value:** Correlation analysis demonstrated that invoice dollar value (Dollars) is the primary determinant of shipping charges (Pearson $r = 0.985$), even exceeding physical unit quantity (Quantity, $r = 0.947$). In carrier logistics, liability insurance, ad-valorem cargo valuation, and bonded shipping surcharges create a strong value-proportional cost baseline.
- **Quantified Economies of Scale:** Exploratory volume quartile analysis revealed a non-linear volume discount structure: low-volume orders (≤ 56 units) incurred a median freight cost of **\$0.10 / unit**, whereas bulk shipments ($\geq 1,008$ units) cost only **\$0.04 / unit**. This represents an empirical **~60% per-unit freight reduction**, proving that linear assumptions alone cannot capture tiered logistics pricing contracts.
- **IQR Outlier Capping:** Raw freight costs exhibited high positive right-skewness (skewness ≈ 3.8). Applying the $1.5 \times \text{IQR}$ capping boundary effectively shielded model gradient updates from extreme enterprise transactions without discarding genuine high-volume orders.
- **Domain Feature Engineering:** Deriving `days_po_to_invoice` (turnaround lead time) and `Price_per_Unit` provided the models with critical density signals, allowing them to differentiate compact high-value shipments from bulky low-margin commodities.

Part 2 Insights: Multi-Model Regression Architecture Evaluation

- **Strong Linear Baseline Performance:** Ordinary Least Squares (OLS) Linear Regression delivered an unexpectedly strong result (**RMSE = \$23.70**, **$R^2 = 0.9863$**). The strong linear correlation between invoice dollars and freight provides a solid performance floor that simplistic non-linear models struggle to beat.
- **Decision Tree Discretization Failure:** A single Decision Tree Regressor underperformed noticeably (**RMSE = \$30.45**, **$R^2 = 0.9774$**). Orthogonal axis-aligned splits partition continuous financial amounts into discrete step-functions, introducing high variance and piecewise prediction jumps on unseen holdout test samples.
- **Ensemble Superiority:** Both Random Forest (bagging) and Gradient Boosting (boosting) successfully overcame single-tree variance. Random Forest reduced error variance by averaging across randomized decorrelated decision trees, while Gradient Boosting systematically corrected residual errors on intermediate price brackets.

Part 3 Insights: Hyperparameter Optimization via GridSearchCV

- **Mitigating Overfitting:** The default Random Forest regressor with unconstrained depth achieved an RMSE of \$25.13 due to leaf-level memorization of supplier noise. Performing `GridSearchCV` with 5-fold cross-validation revealed that constraining tree depth to `max_depth=6` and setting `min_samples_split=5` produced the project's best test score: **RMSE = \$23.46** and **$R^2 = 0.9866$** .
- **Ensemble Parsimony:** An ensemble of 50 constrained trees outperformed deeper 100-tree models, demonstrating that in tabular data with 4 well-engineered features, shallower trees prevent variance accumulation while fully capturing non-linear bulk thresholds.
- **Cross-Validated Generalization:** 5-fold cross-validation ensured that hyperparameter tuning was immune to holdout test set anomalies, guaranteeing high predictive stability across diverse supplier categories.

Part 4 Insights: Streamlit Web Deployment & Operational Business ROI

- **Bridging Data Science to Operational Workflows:** Deploying the models into an interactive web dashboard at <https://adsfa2-machine-learning.streamlit.app/> transforms static academic models into an actionable tool for procurement agents and finance managers.
- **Multi-Model Consensus Transparency:** Presenting predictions from all 5 models side-by-side provides procurement teams with a clear confidence band, dispelling black-box skepticism and validating estimates against algorithmic consensus.
- **Automated Invoice Auditing & Margin Protection:** In commercial retail logistics, carrier overbilling represents a persistent source of margin leakage. Incorporating automated freight sanity thresholds (>15% deviation from expected freight proportion) empowers finance departments to flag and dispute incorrect vendor invoices before disbursement.
- **Production Pipeline Consistency:** Serializing both the model and the exact fitted `StandardScaler` guarantees zero training-serving skew, ensuring that live inference calculations match training conditions precisely.